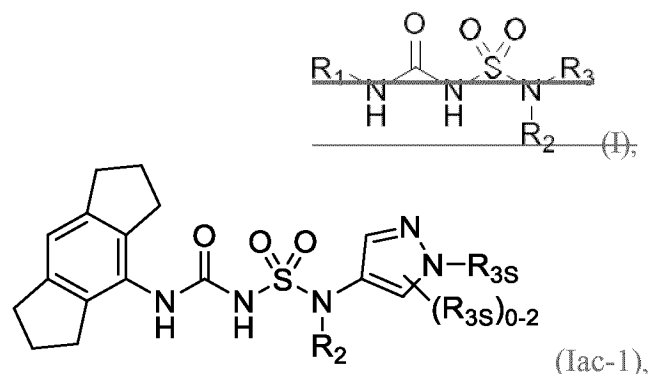


MAIN REQUEST

CLAIMS:

1. _____ 1. ——— A compound of Formula (Iac1):



or a solvate, or pharmaceutically acceptable salt thereof, wherein:

~~R₁ is C₃-C₁₆ cycloalkyl optionally substituted by one or more R_{1S}, wherein the C₃-C₁₆ cycloalkyl is a C₃-C₈ monocyclic cycloalkyl or polycyclic cycloalkyl, or, R₁ is a phenyl substituted by two or three R_{1S}, wherein each R_{1S} is independently C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₁-C₆ alkoxy, C₁-C₆ haloalkoxy, or halo, and at least one R_{1S} is C₁-C₆ alkyl, C₁-C₆ haloalkyl, or halo;~~

~~R₂ is -(CHX₂X₂)_n-R_{2S}, wherein n is 0, 1, or 2, and each X₂ is independently H, C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl, wherein the C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl is optionally substituted with one or more halo, CN, OH, O(C₁-C₆ alkyl), NH₂, NH(C₁-C₆ alkyl), N(C₁-C₆ alkyl)₂, or oxo;~~

~~R_{2S} is 4- to 8-membered monocyclic heterocycloalkyl optionally substituted with one or more C₁-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₁-C₆ haloalkyl, halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo; and~~

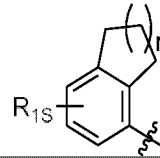
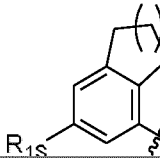
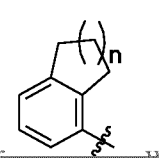
~~R₃ is 7- to 12-membered heterocycloalkyl or 5- or 6-membered heteroaryl optionally substituted with one or more R_{3S}, wherein each R_{3S} is independently C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₃-C₈ heterocycloalkyl wherein the C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with -O(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, halo, or -CN.~~

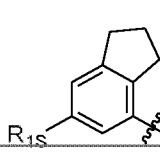
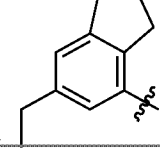
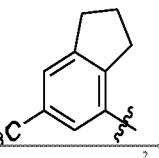
2. — The compound of claim 1, wherein R_1 is C_3 – C_7 monocyclic cycloalkyl, C_9 – C_{10} bicyclic cycloalkyl, C_{12} – C_{16} tricyclic cycloalkyl, or phenyl, wherein the C_3 – C_7 monocyclic cycloalkyl, C_9 – C_{10} bicyclic cycloalkyl, or C_{12} – C_{16} tricyclic cycloalkyl is optionally substituted by one or more R_{1s} , the phenyl is substituted by two or three R_{1s} ; and wherein at least one R_{1s} is C_1 – C_6 alkyl, C_1 – C_6 haloalkyl, or halo;

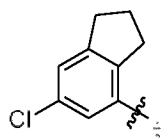
optionally wherein:

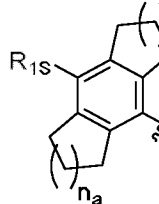
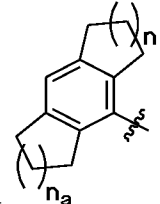
(a) the C_3 – C_7 monocyclic cycloalkyl is cyclopentyl, cyclohexyl, or cycloheptyl, wherein the cyclopentyl, cyclohexyl, or cycloheptyl is optionally substituted by one or more R_{1s} ; further

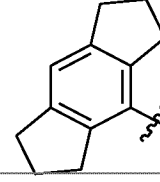


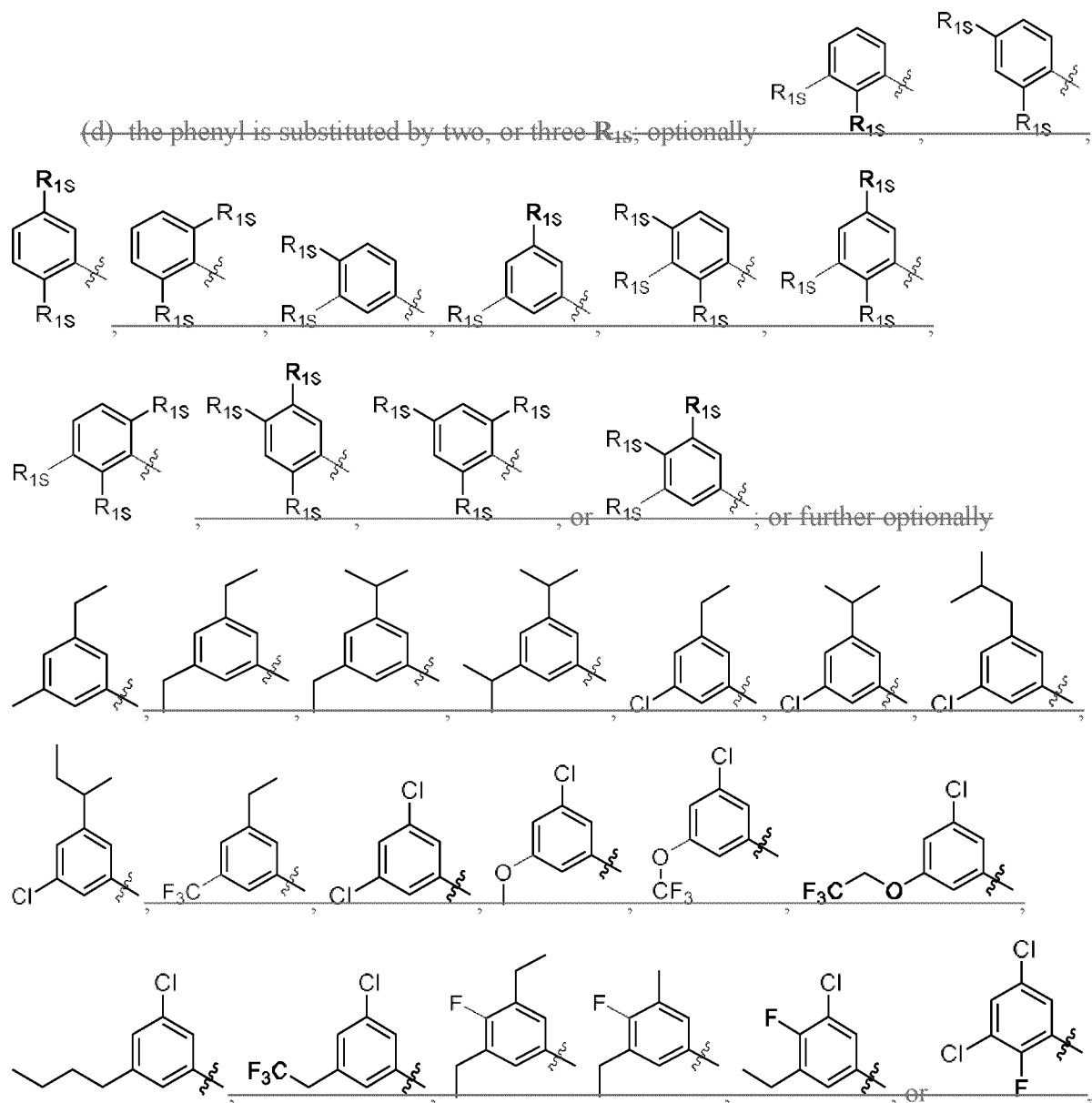
(b) the C_9 – C_{10} bicyclic cycloalkyl is , , or , wherein n

is 0, 1, or 2; optionally ; further optionally , , or



(c) the C_{12} – C_{16} tricyclic cycloalkyl is  or , wherein n and n_a

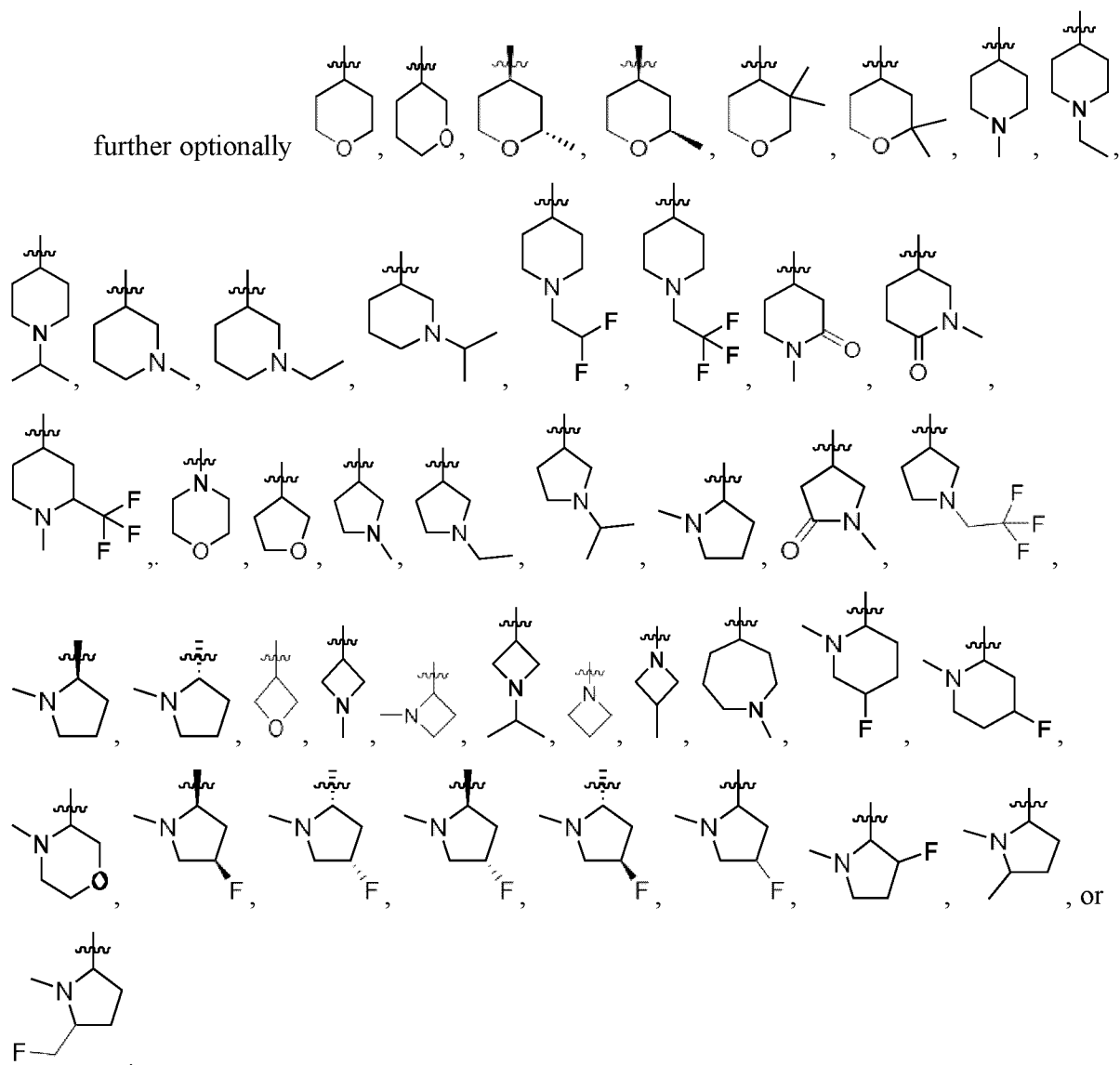
each independently are 0, 1, 2, or 3; further optionally ; and/or



3. ~~The compound of claim 1 or 2, wherein R_2 is R_{2s} , $(CX_2X_2)R_{2s}$, or $(CX_2X_2)_2R_{2s}$.~~

42. ~~The compound of any one of claims 1 to 3, wherein R_{2s} is 4- to 8-membered monocyclic heterocycloalkyl optionally substituted with one or more C_1 - C_6 alkyl, C_2 - C_6 alkenyl, C_2 - C_6 alkynyl, C_1 - C_6 haloalkyl, halo, CN, OH, $O(C_1$ - C_6 alkyl), NH_2 , $NH(C_1$ - C_6 alkyl), $N(C_1$ - C_6 alkyl) $_2$, or oxo;~~

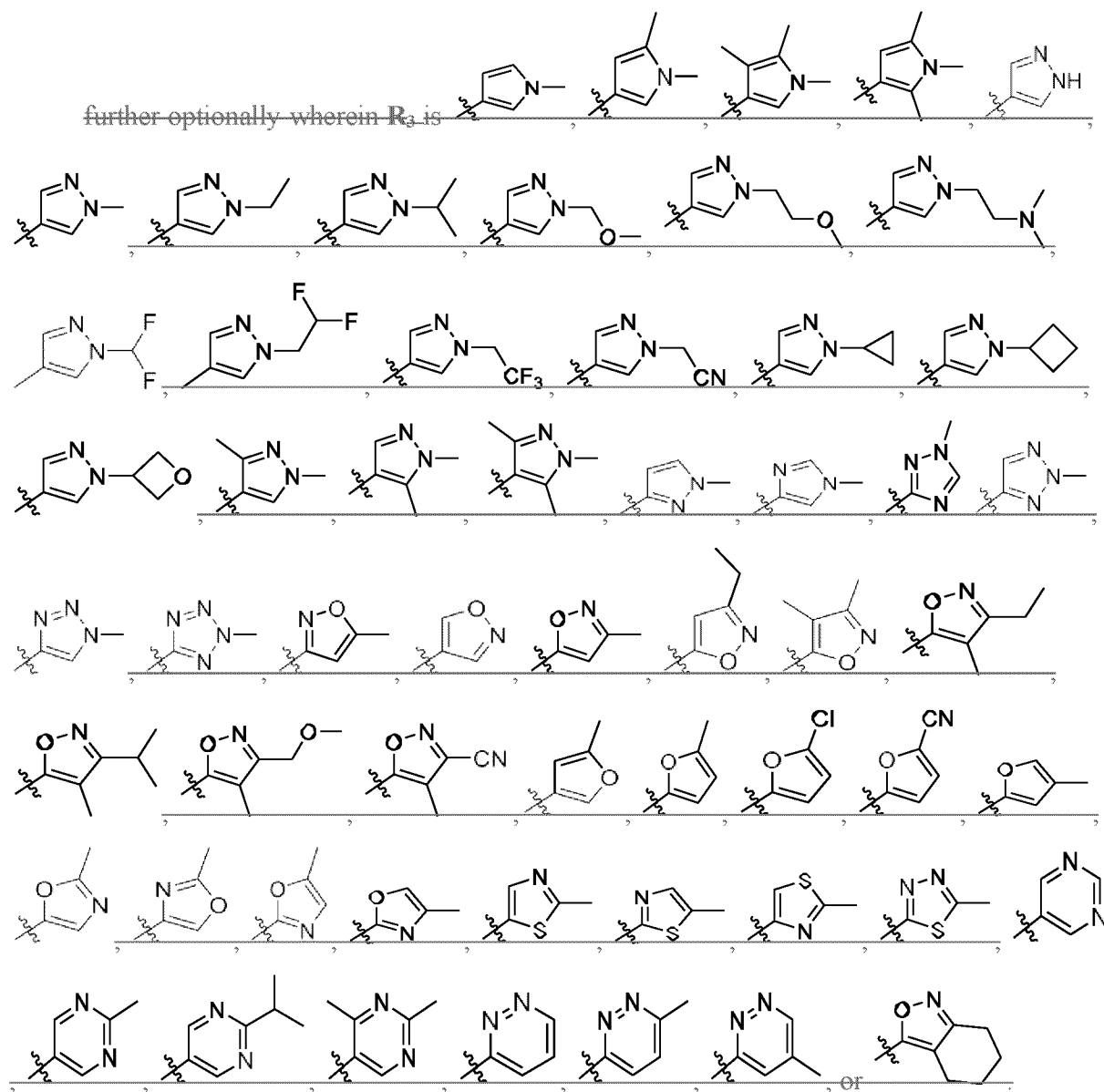
Optionally oxetanyl, azetidiny, tetrahydrofuranyl, pyrrolidinyl, tetrahydropyranyl, piperidinyl, morpholinyl, or azepanyl, wherein the oxetanyl, azetidiny, tetrahydrofuranyl, pyrrolidinyl, tetrahydropyranyl, piperidinyl, morpholinyl, or azepanyl is optionally substituted with one or more C₁-C₆ alkyl, C₁-C₆ haloalkyl, or oxo; or



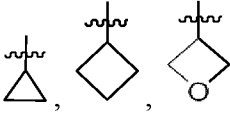
5. The compound of any one of claims 1 to 4, wherein **R₃** is 7- to 12-membered heterocycloalkyl optionally substituted with one or more **R_{3s}**;

optionally wherein **R₃** is pyrrolyl, pyrazolyl, imidazolyl, triazolyl, tetrazolyl, isoxazolyl, furanyl, oxazolyl, isothiazolyl, thiadiazolyl, pyridinyl, pyrimidinyl, pyridazinyl, or 4,5,6,7-

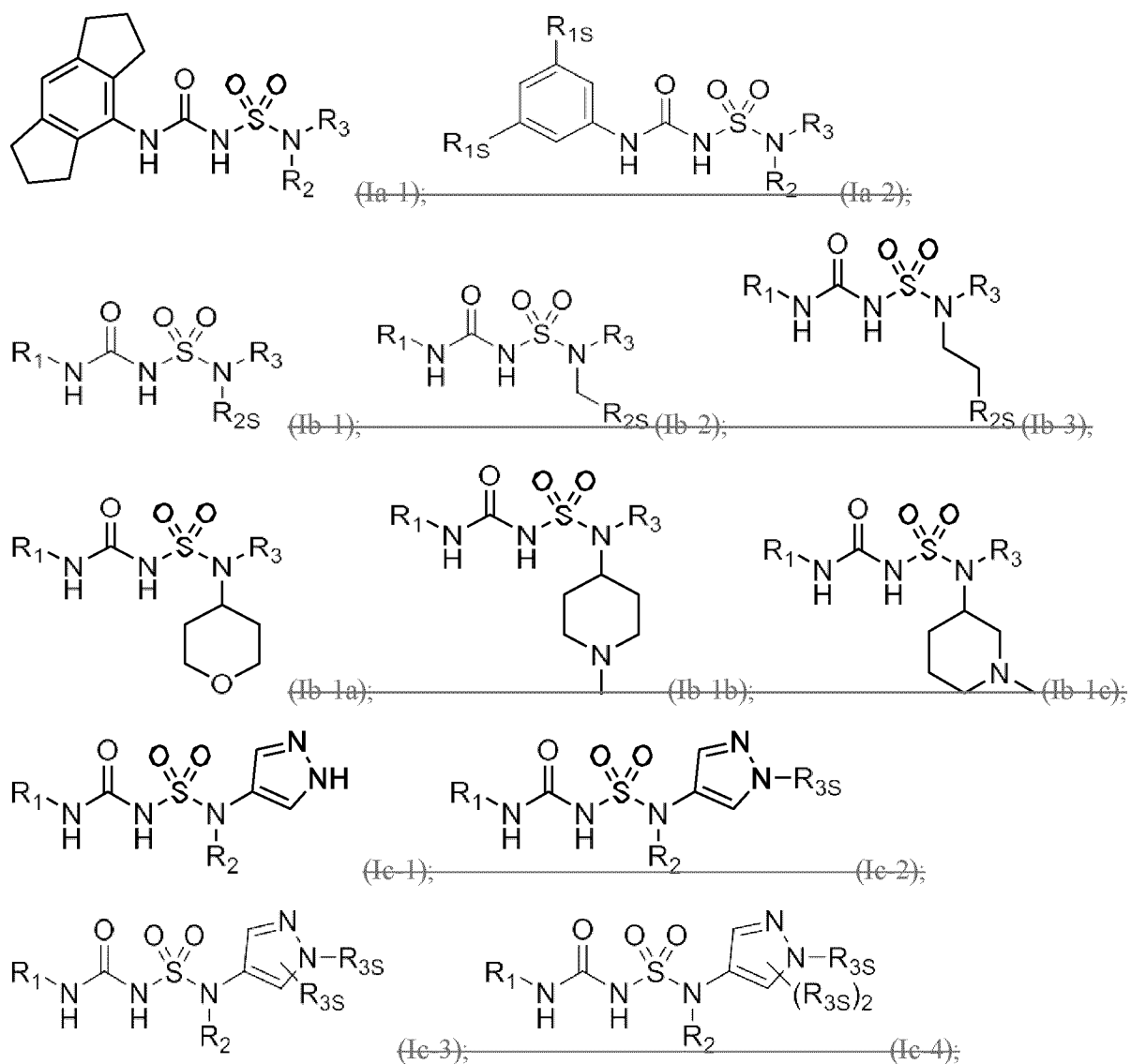
~~tetrahydrobenzo[c]isoxazole, wherein the pyrrolyl, pyrazolyl, imidazolyl, triazolyl, tetrazolyl, isoxazolyl, furanyl, oxazolyl, isothiazolyl, thiadiazolyl, pyridinyl, pyrimidinyl, pyridazinyl, or 4,5,6,7-tetrahydrobenzo[c]isoxazole is optionally substituted with one or more R_{35} ; or~~

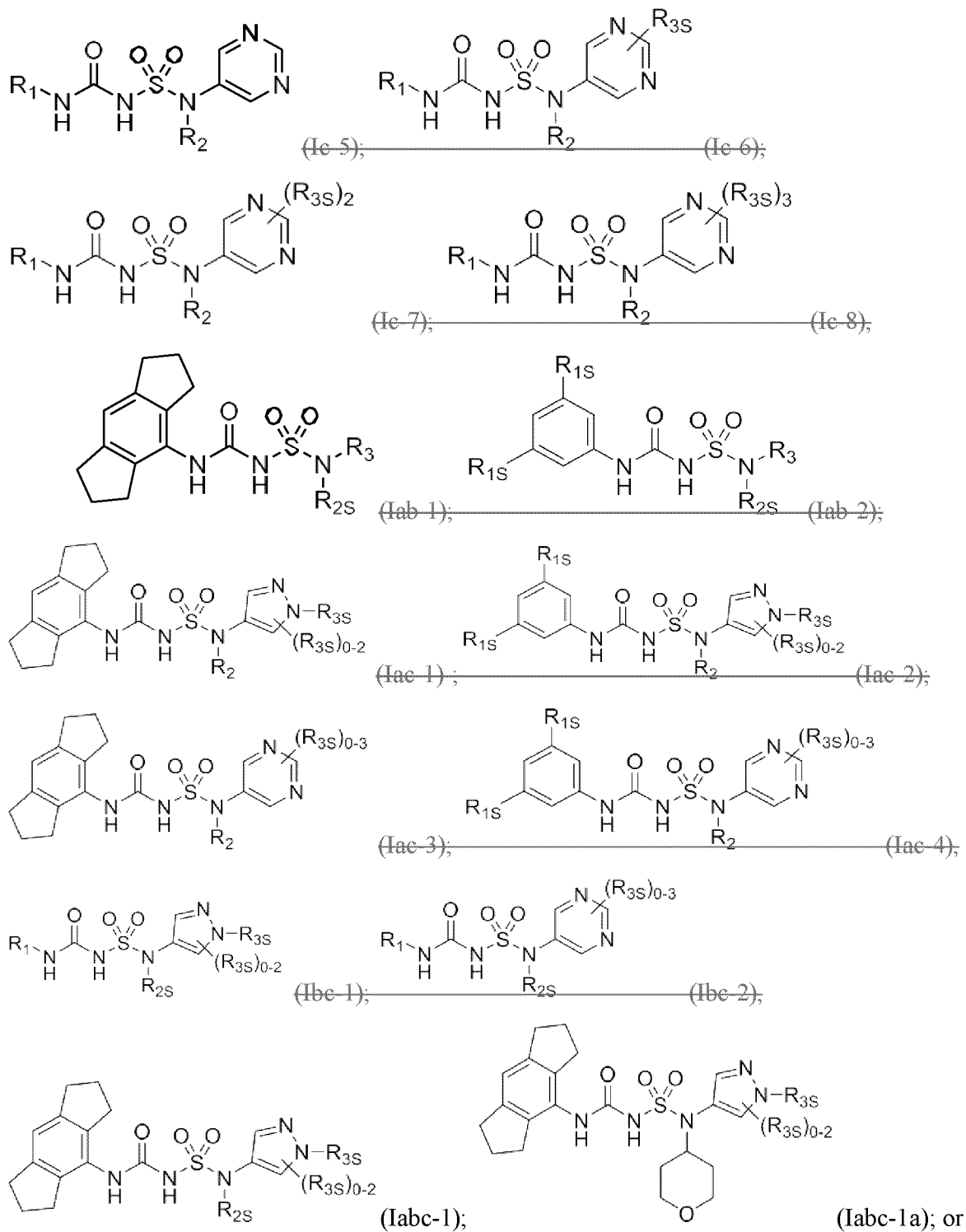


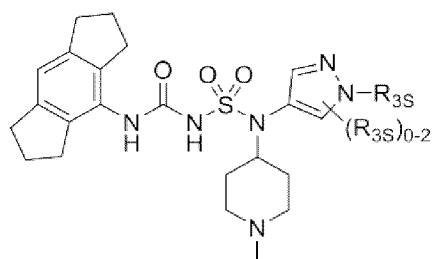
~~36. The compound of any one of claims 1 or claim 2 to 5, wherein at least one R_{35} is C_1 - C_6 alkyl, C_1 - C_6 haloalkyl, C_3 - C_8 cycloalkyl, halo, or C_3 - C_8 heterocycloalkyl wherein the C_1 - C_6 alkyl, C_1 - C_6 haloalkyl, C_3 - C_8 cycloalkyl, or C_3 - C_8 heterocycloalkyl is optionally substituted with $O(C_1$ - C_6 alkyl), $N(C_1$ - C_6 alkyl)₂, halo, or CN;~~

optionally wherein at least one R_{3s} is , methyl, ethyl, isopropyl, -CH₂OCH₃, -CH₂CF₃, -CH₂CH₂OCH₃, -CH₂CN, CH₂CF₂, or -CH₂CH₂N(CH₃)₂.

74. The compound of any one of the preceding claims, wherein the compound is of Formula (Ia-1), (Ia-2), (Ib-1), (Ib-2), (Ib-3), (Ib-1a), (Ib-1b), (Ib-1c), (Ic-1), (Ic-2), (Ic-3), (Ic-4), (Ic-5), (Ic-6), (Ic-7), (Ic-8), (Iab-1), (Iab-2), (Iac-1), (Iac-2), (Iac-3), (Iac-4), (Ibc-1), (Ibc-2), (Iabc-1), (Iabc-1a), or (Iabc-1b):



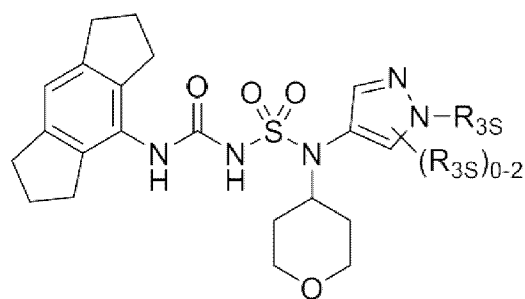




(Iabc-1b)

or a solvate, or pharmaceutically acceptable salt thereof, wherein R_1 , R_2 , R_3 , and R_{3S} are as described ~~herein~~ in claim 1.

~~85.~~ The compound of any one of claims 1 to ~~74~~, wherein the compound is of Formula (Iabc-1a):

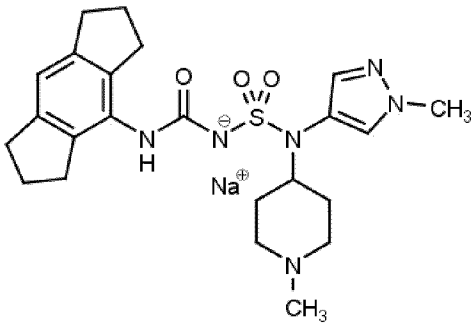
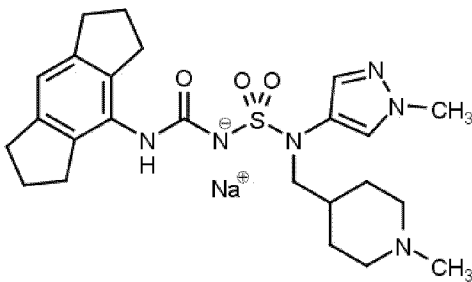
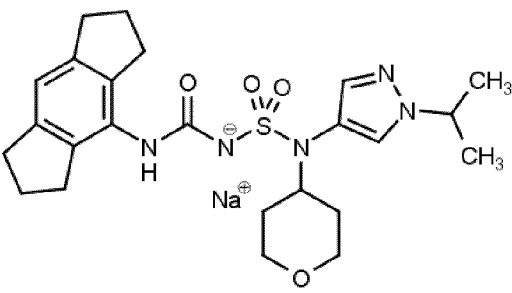
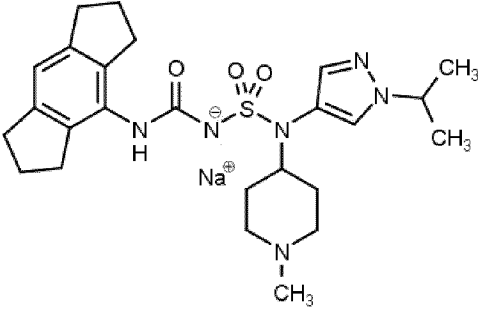


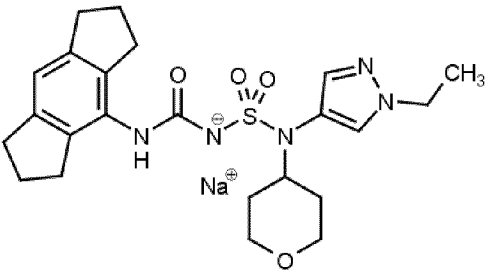
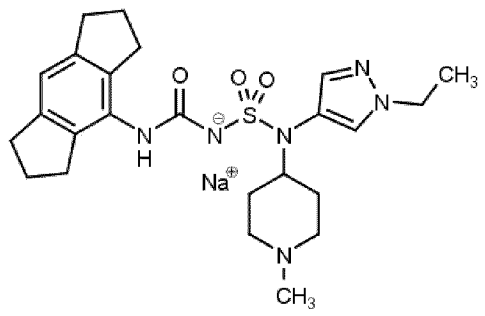
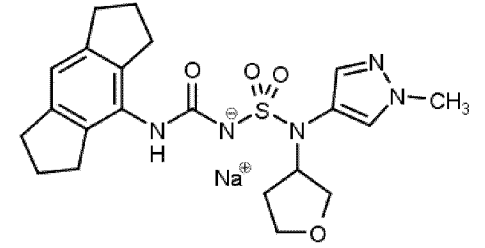
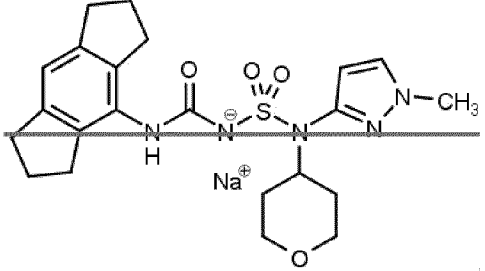
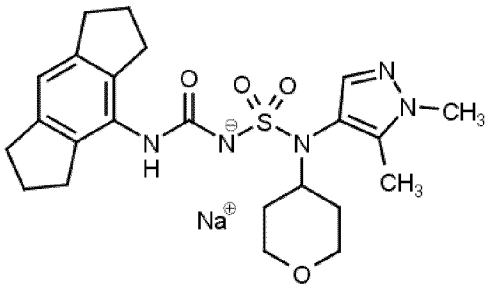
(Iabc-1a)

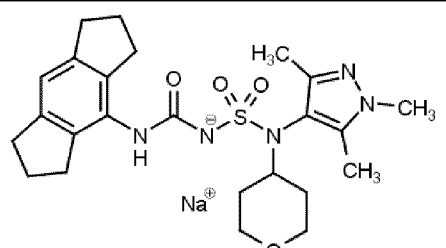
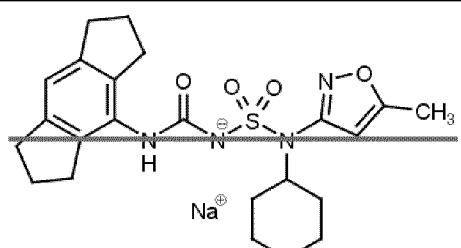
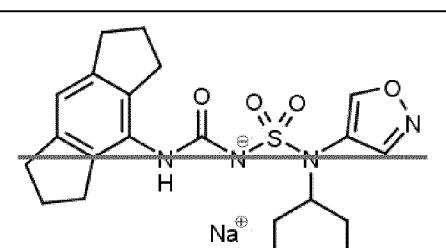
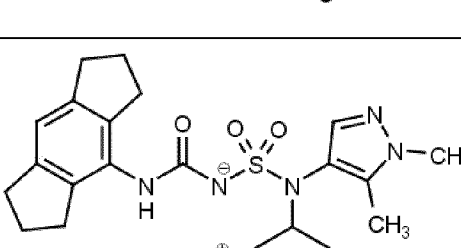
or a solvate, or pharmaceutically acceptable salt thereof, wherein R_{3S} is as described ~~herein~~ in claim 1.

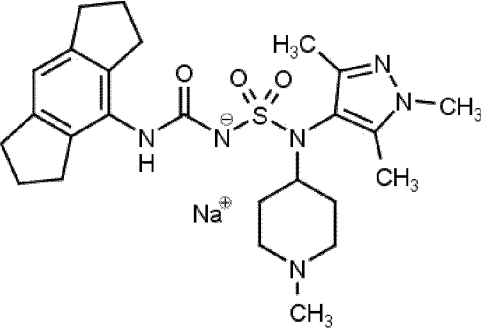
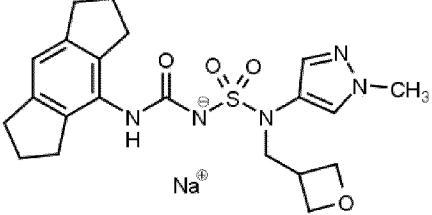
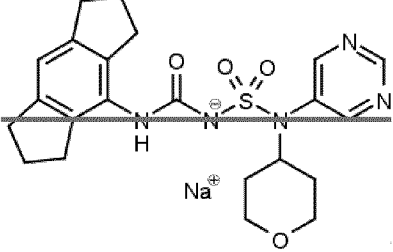
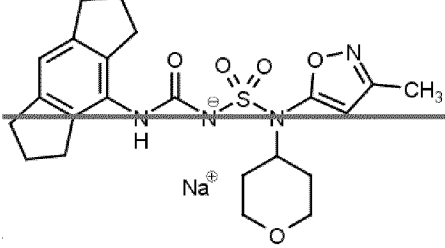
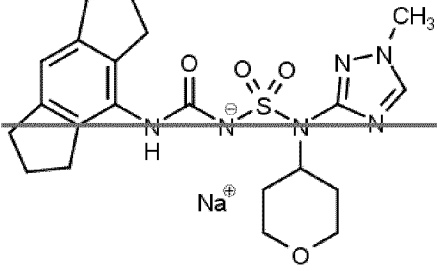
~~96.~~ The compound of any one of ~~the~~ claims 1 to ~~85~~, being selected from:

Compound No.	Structure
1	

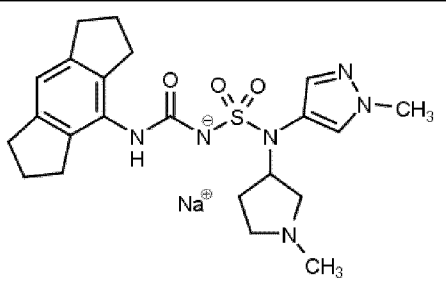
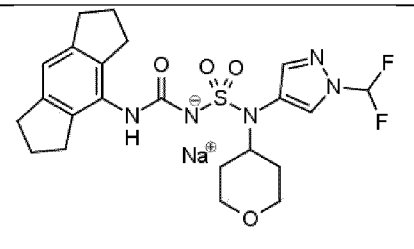
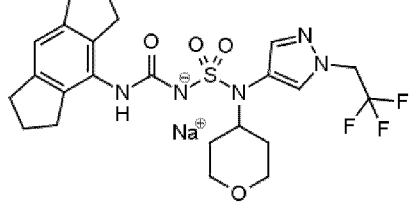
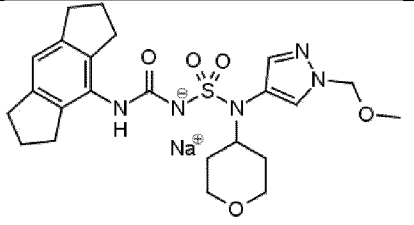
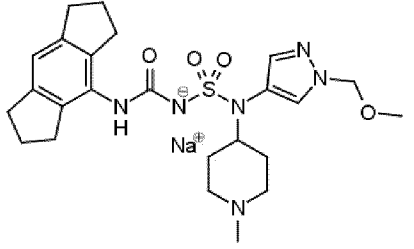
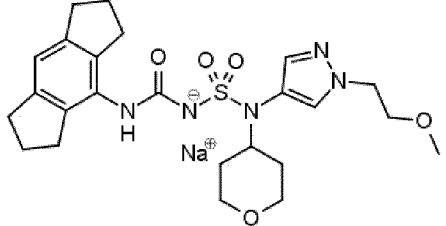
Compound No.	Structure
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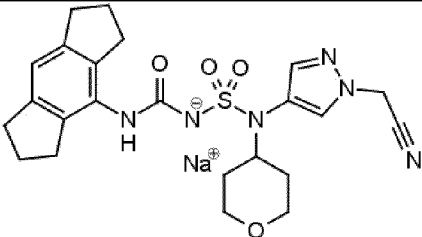
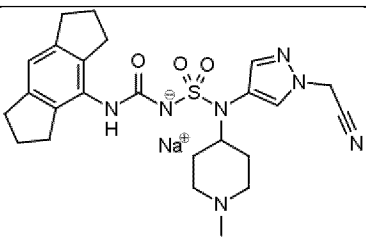
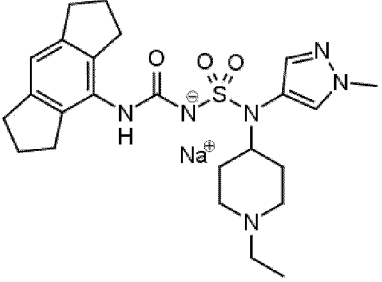
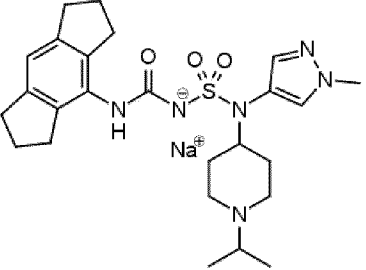
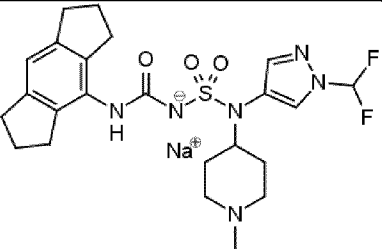
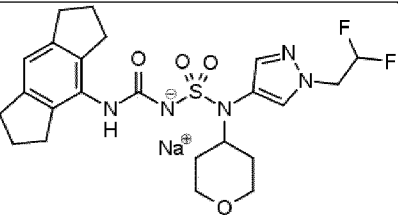
Compound No.	Structure
6	
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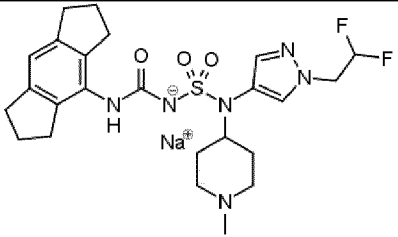
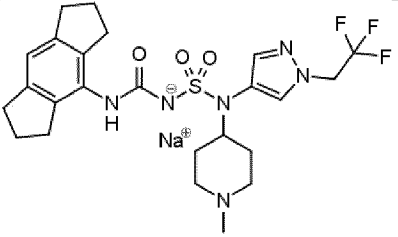
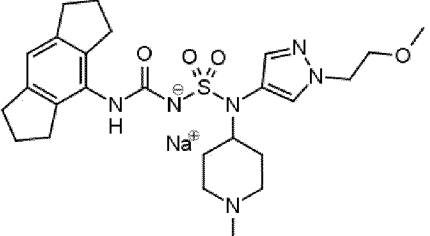
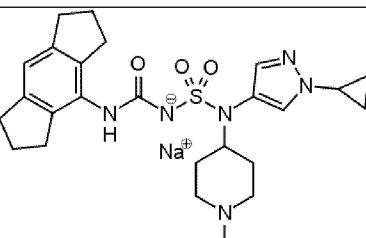
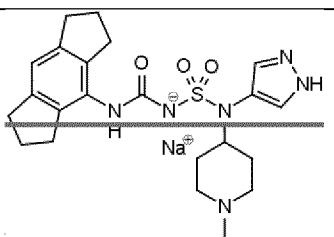
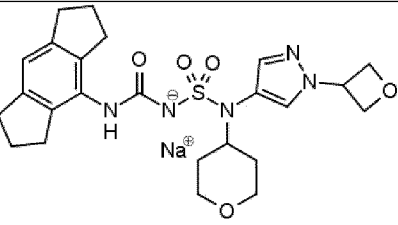
Compound No.	Structure
11	
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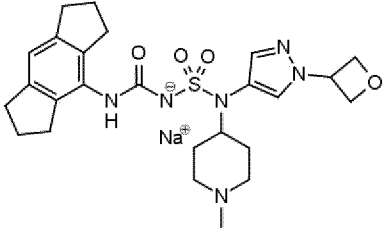
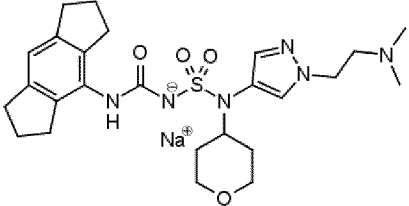
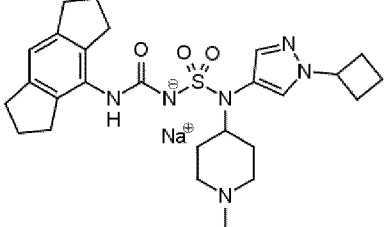
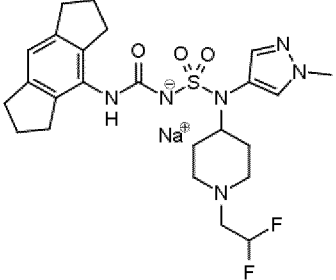
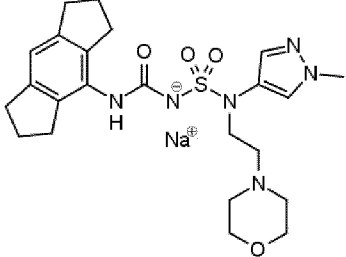
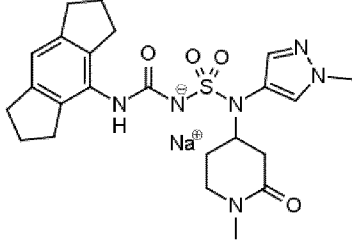
Compound No.	Structure
15	
16	
17	
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19	

Compound No.	Structure
20	 <chem>CN1CCN(C1)C(=O)NS(=O)(=O)N2CCCCC2.[Na+]</chem>
21	 <chem>CN1CCN(C1)C(=O)NS(=O)(=O)N2CCOCC2.[Na+]</chem>
22	 <chem>CN1CCN(C1)C(=O)NS(=O)(=O)N2CCOCC2.[Na+]</chem>
23	 <chem>CN1CCN(C1)C(=O)NS(=O)(=O)N2CCCCC2.[Na+]</chem>

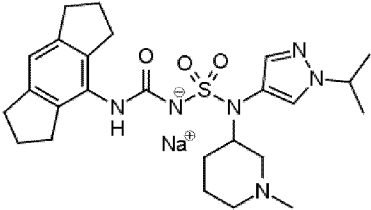
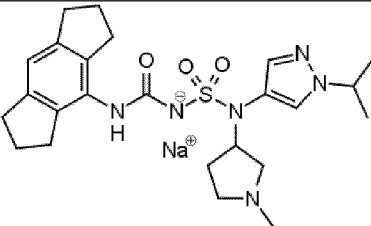
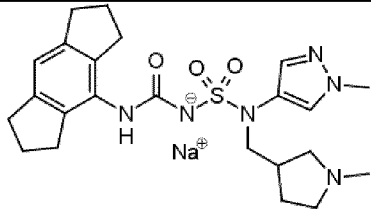
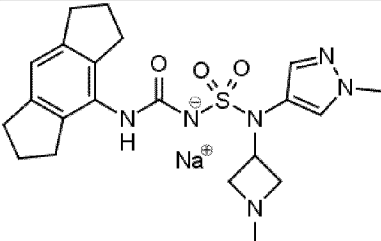
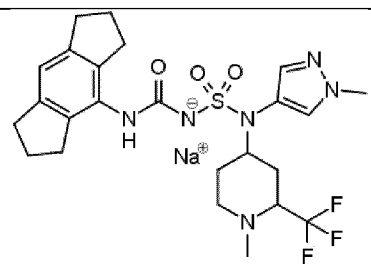
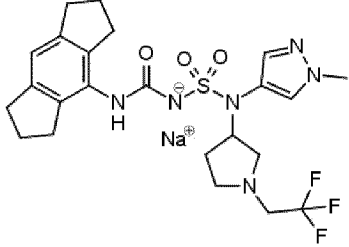
Compound No.	Structure
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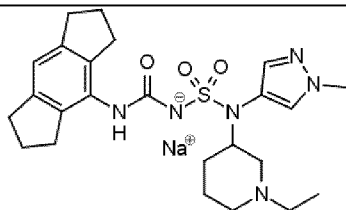
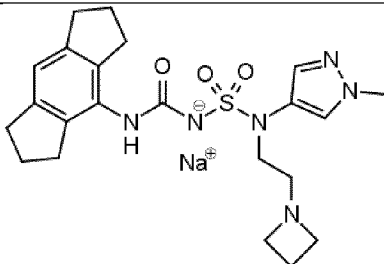
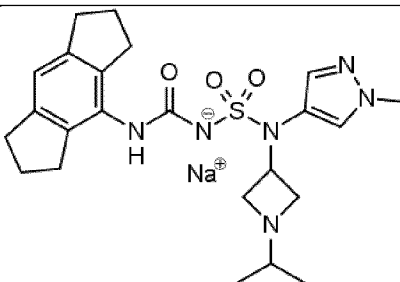
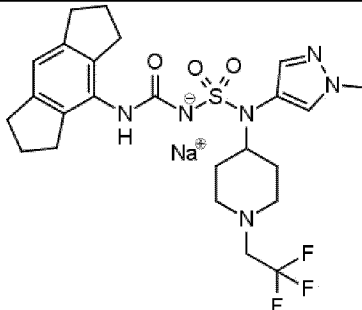
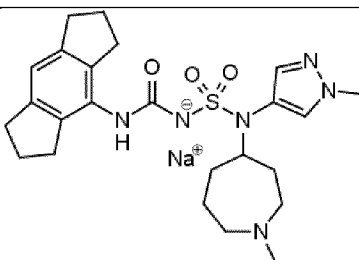
Compound No.	Structure
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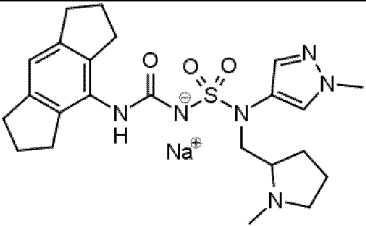
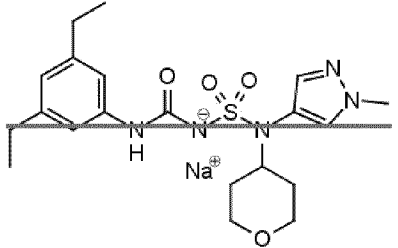
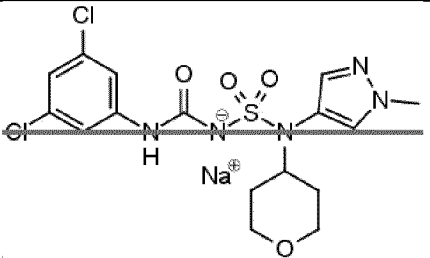
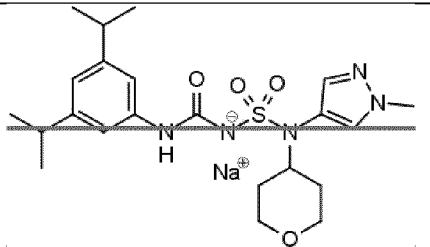
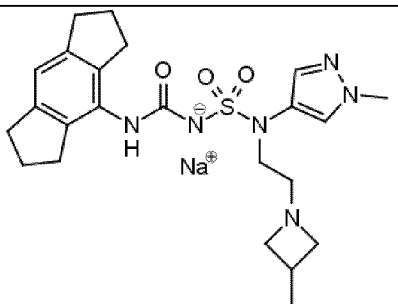
Compound No.	Structure
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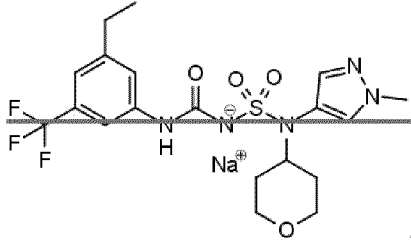
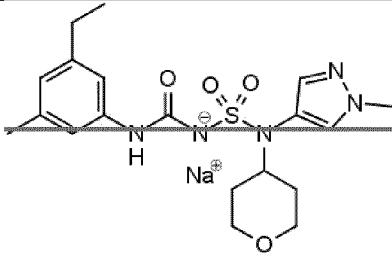
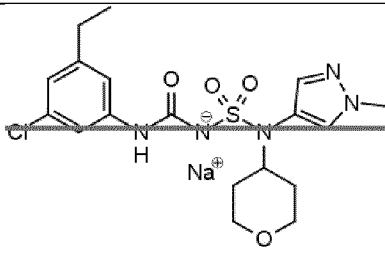
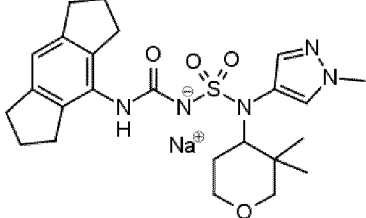
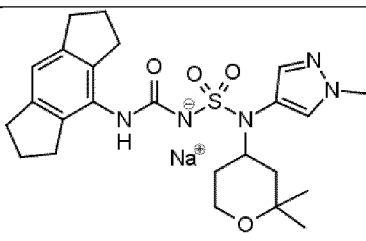
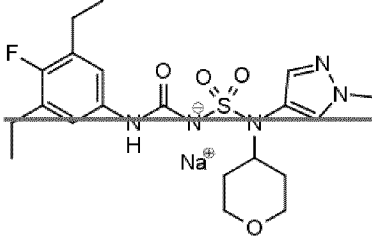
Compound No.	Structure
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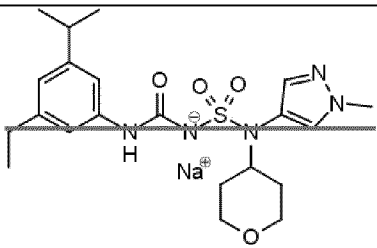
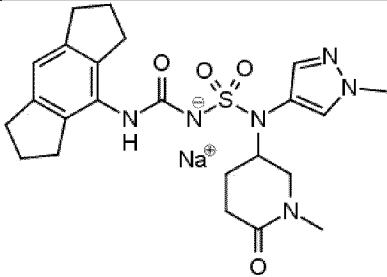
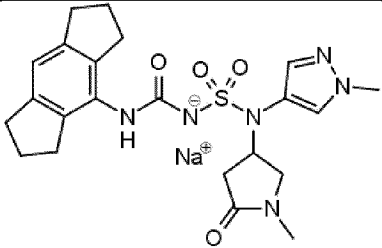
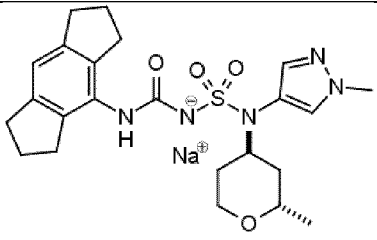
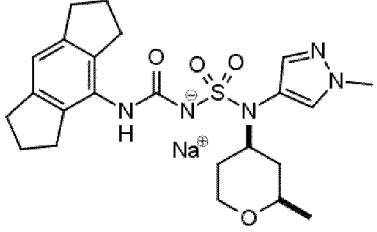
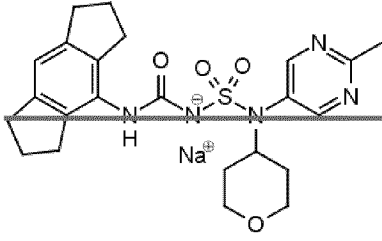
Compound No.	Structure
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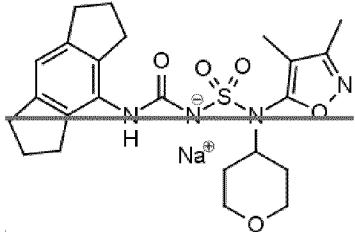
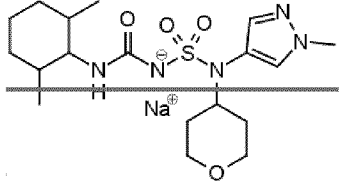
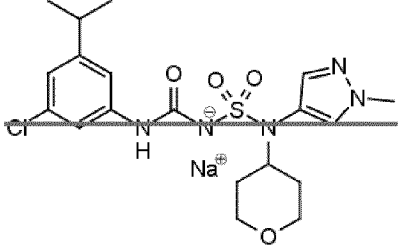
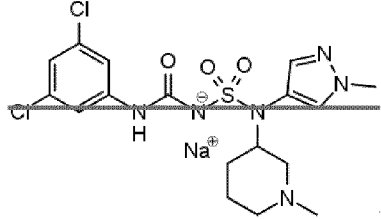
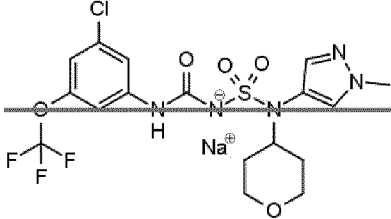
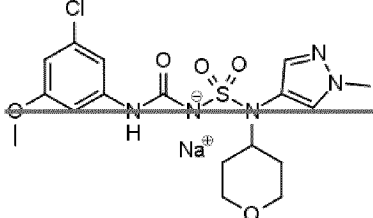
Compound No.	Structure
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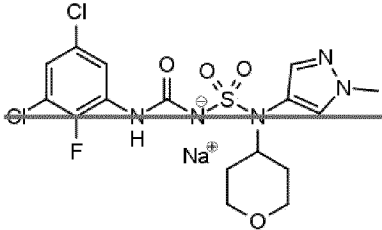
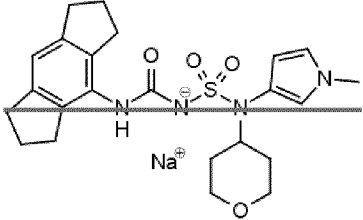
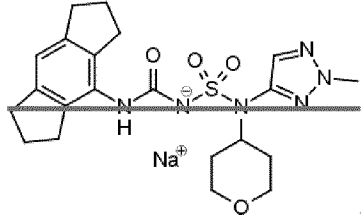
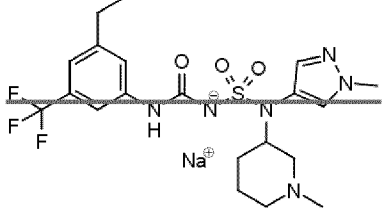
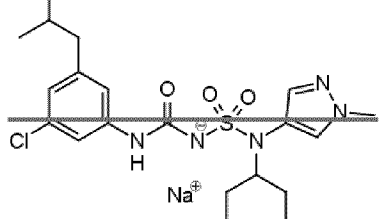
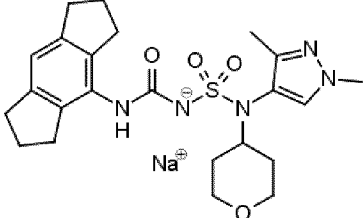
Compound No.	Structure
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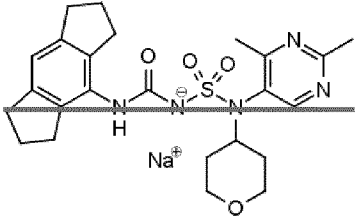
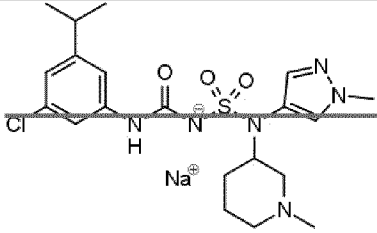
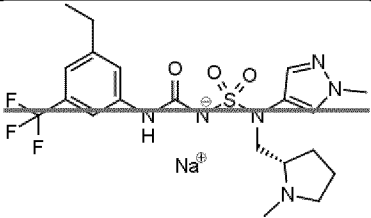
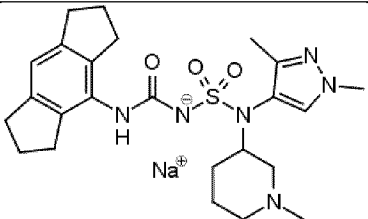
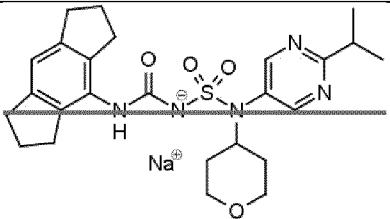
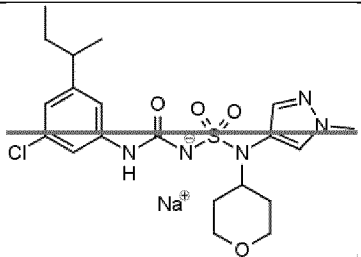
Compound No.	Structure
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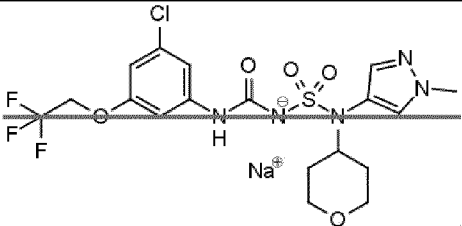
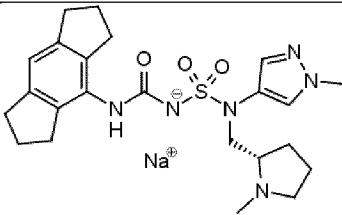
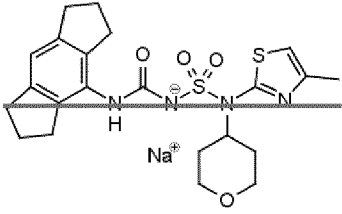
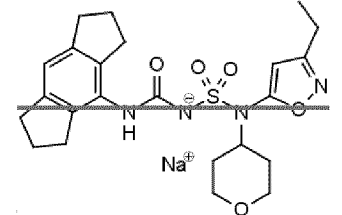
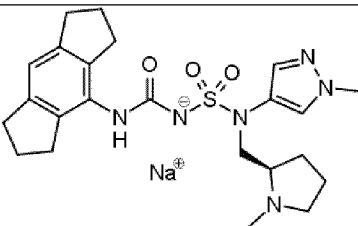
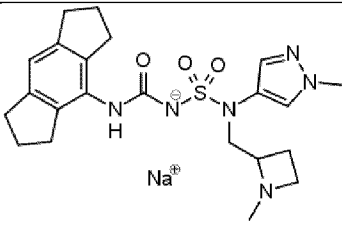
Compound No.	Structure
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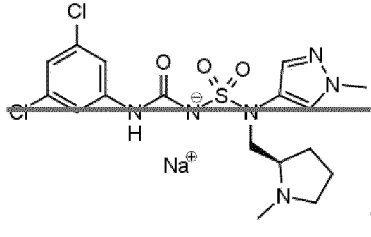
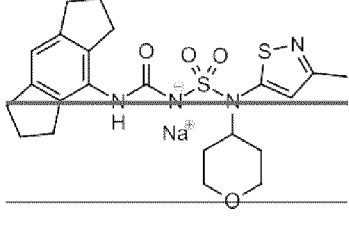
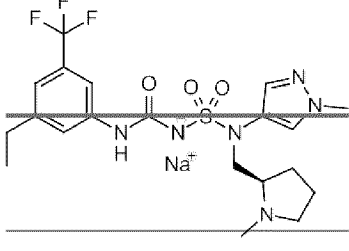
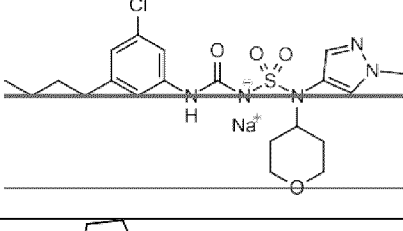
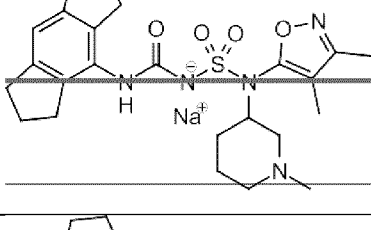
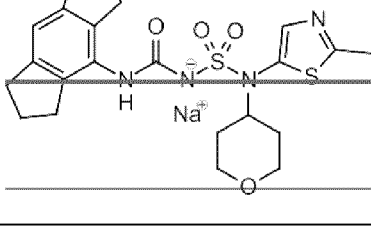
Compound No.	Structure
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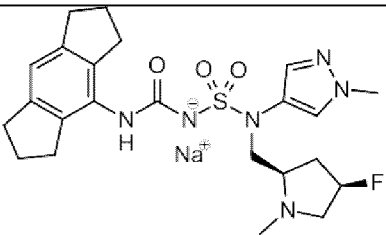
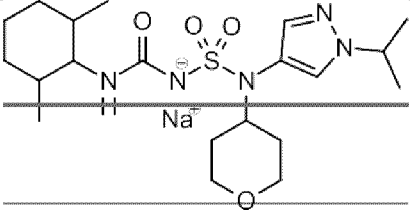
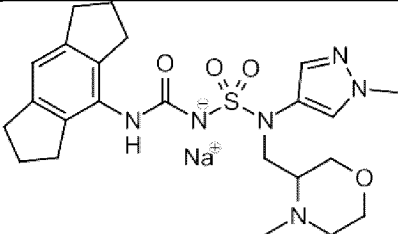
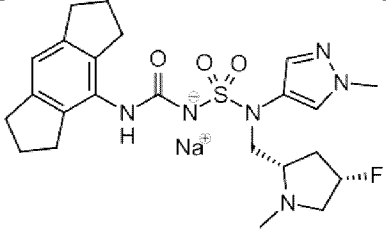
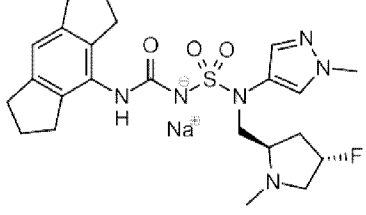
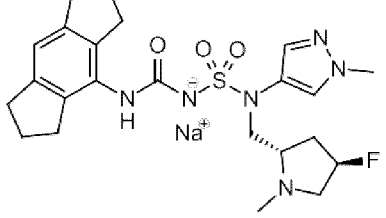
Compound No.	Structure
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Compound No.	Structure
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Compound No.	Structure
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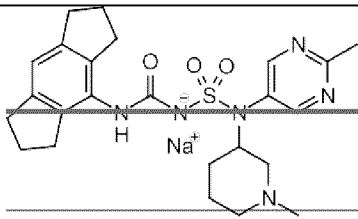
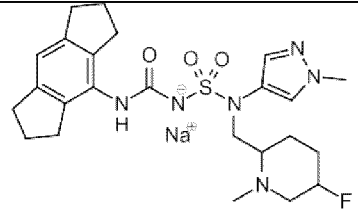
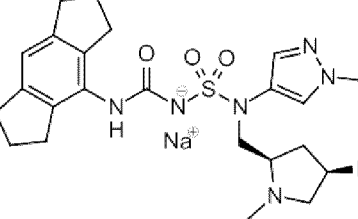
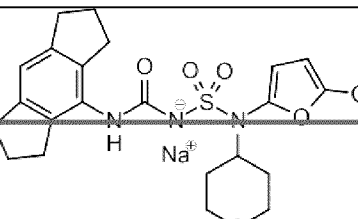
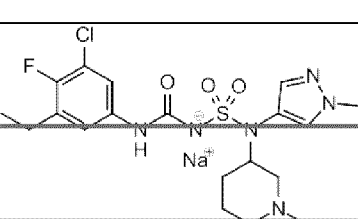
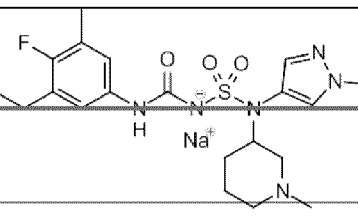
Compound No.	Structure
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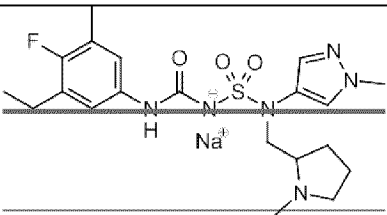
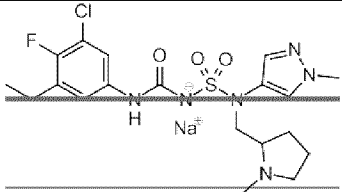
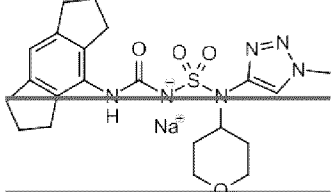
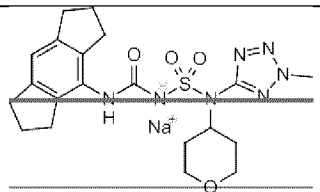
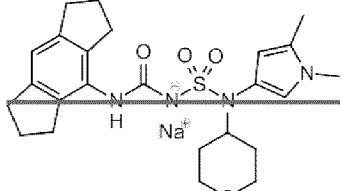
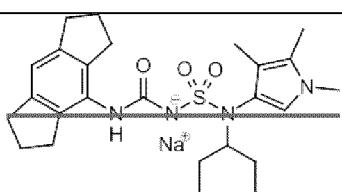
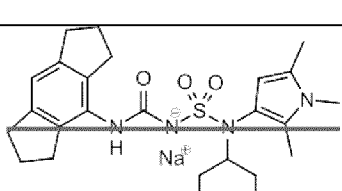
Compound No.	Structure
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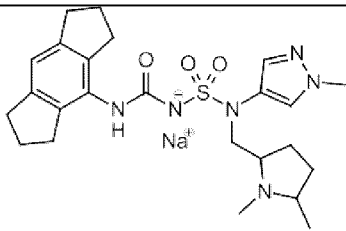
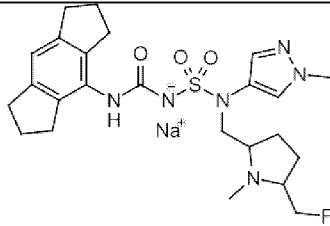
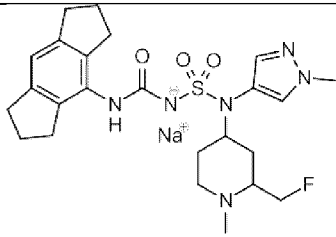
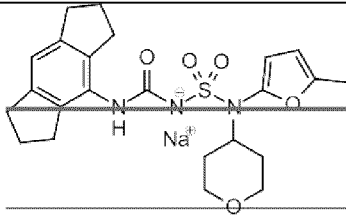
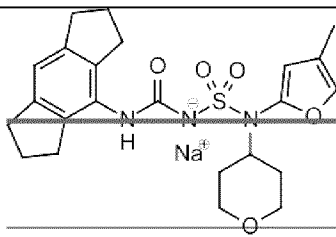
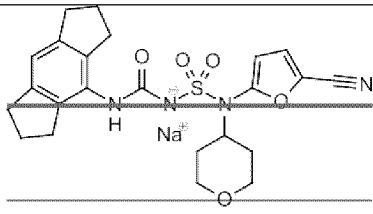
Compound No.	Structure
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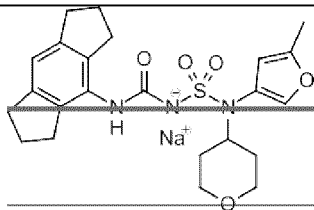
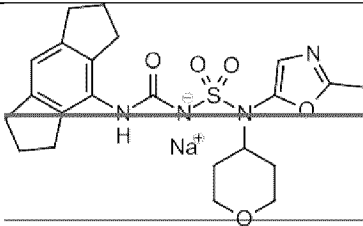
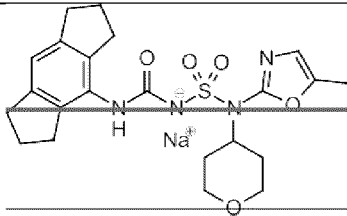
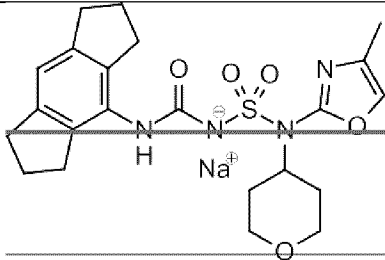
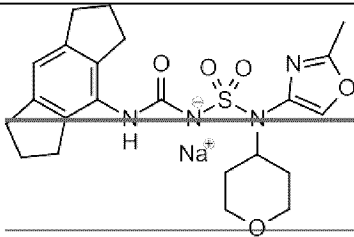
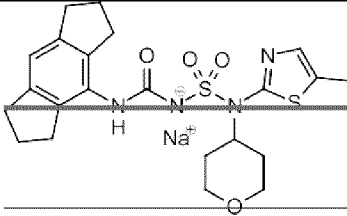
Compound No.	Structure
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Compound No.	Structure
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Compound No.	Structure
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Compound No.	Structure
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Compound No.	Structure
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Compound No.	Structure
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Compound No.	Structure
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and pharmaceutically acceptable salts thereof.

~~107~~. A pharmaceutical composition comprising the compound of any one of claims 1 to ~~96~~ or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable diluent or carrier.

~~118~~. The compound of any one of claims 1 to ~~96~~, or the pharmaceutical composition of claim ~~107~~,

for use in inhibiting inflammasome activity; optionally, the inflammasome is NLRP3 inflammasome, and the activity is *in vitro* or *in vivo*.

~~129~~. The compound of any one of claims 1 to ~~96~~, or the pharmaceutical composition of claim ~~107~~, for use in treating or preventing a disease or disorder; preferably wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder, an autoimmune disorder, a neurodegenerative disease, or cancer.

~~1310~~. The compound or pharmaceutical composition for use of claim ~~118~~ or claim 129, wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder or an autoimmune disorder; optionally, the disease or disorder is selected from cryopyrin-associated autoinflammatory syndrome (CAPS; e.g., familial cold autoinflammatory syndrome (FCAS), Muckle-Wells syndrome (MWS), chronic infantile neurological cutaneous and articular (CINCA) syndrome/ neonatal-onset multisystem inflammatory disease (NOMID)), familial Mediterranean fever (FMF), nonalcoholic fatty liver disease (NAFLD), non-alcoholic steatohepatitis (NASH), gout, rheumatoid arthritis, osteoarthritis, Crohn's disease, chronic obstructive pulmonary disease (COPD), chronic kidney disease (CKD), fibrosis, obesity, type 2 diabetes, multiple sclerosis, dermatological diseases (e.g., acne) and neuroinflammation occurring in protein misfolding diseases (e.g., Prion diseases).

~~1411~~. The compound or pharmaceutical composition for use of claims ~~118~~ or claim 129, wherein disease or disorder is a neurodegenerative disease; optionally, the disease or disorder is Parkinson's disease or Alzheimer's disease.

~~1512~~. The compound or pharmaceutical composition for use of claims ~~118~~ or claim 129, wherein the disease or disorder is cancer; optionally, the cancer is metastasising cancer, gastrointestinal cancer, brain cancer, skin cancer, non-small-cell lung carcinoma, head and neck squamous cell carcinoma or colorectal adenocarcinoma.